

PythonLearning-2026

🌟 Key highlights (关键亮点)



A comprehensive Python learning repository! Welcome!

这是一个包含了“基础”、“进阶”、“标准库”和“实战”的 **python** 学习仓库



Free Teaching Videos: (最新教学视频-完全免费)

<https://space.bilibili.com/3690991649294439/lists/7284550>

<https://xianzs.github.io/>



Get Started Quickly (快速上手)

You need to make sure that the git tool has been installed on your computer.



```
## 1.use git tool
```

```
git clone
https://github.com/XianZS/PythonLearning
## 2.Configuration Environment (either-
or)
cd PythonLearning
### 2.1 if you use conda
conda env create -f environment.yml
### 2.2 if you use pip
pip install -r requirements.txt
## 3.Export environment
conda env export > environment.yml
pip freeze > requirements.txt
```



Feedback exchange（交流反馈）

Welcome everyone's participation in the discussion. Thank you.（欢迎各位的交流，谢谢。）

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Module Order / Complete Directory Structure（项目架构）



PythonLearning/

└─ Python Basic Learning Section/

[python 基础学习]

| └─ 1. Getting Started Guide

(Icebreaker)

| └─ 2. Variables and Data Types

| └─ 3. Operators and Expressions

| └─ 4. Flow Control (Core Foundation)

| └─ 5. Function Basics

| └─ 6. Data Structures

(List/Tuple/Dictionary/Set)

| └─ 7. File Operations

| └─ 8. Exception Handling

| └─ 9. Basic Comprehensive Practical

Project

└─ Python Advanced Learning Section/

[python 进阶学习]

| └─ 1. Advanced Functions

| └─ 2. Object-Oriented Programming

(OOP)

| └─ 3. Modularization and Package

Management

| └─ 4. Basic Data Processing

(Connecting Standard Libraries/Third-

Party Libraries)

- | └─ 5. Basic Network Programming

- | └─ 6. Database Programming

- | └─ 7. Introduction to Concurrent Programming

- | └─ 8. Advanced Comprehensive Practical Projects (2 Projects, Covering Multiple Modules)

- | └─ 9. Advanced Stage Review and Direction Guidance

- └─ Python Standard Library Learning Section/ [python 标准库学习]

- └─ 1. File and Directory Operations

- └─ 2. Data Processing and Formats

- └─ 3. System Interaction and

Processes

- └─ 4. Text Processing

- └─ 5. Date and Time

- └─ 6. Mathematics and Scientific

Computing

- └─ 7. Concurrency and Asynchrony

- └─ 8. Debugging and Testing

- └─ 9. Network Communication

- └─ 10. Encryption and Security



document description (学习文档详细描述)

1.1 Python Basic Learning Section

Module Order	Core Content
1. Getting Started Guide (Icebreaker)	① What is Python/What can it do (Visual demonstration of application scenarios such as web crawling, data analysis, automation, etc.); ② Environment Setup (Adapted for Windows/Mac/Linux systems, Anaconda+PyCharm installation, solving the pain point of "installation errors"); ③ First Program (Hello World + personalized modification, such as outputting your own name to enhance sense of involvement)
2. Variables and Data Types	① Variable Definition (Naming rules + taboos, pitfalls to avoid: Chinese variables, keyword naming); ② Basic Data Types (int/float/str/bool, each type with 2 life-oriented cases: e.g., str for processing names, int for calculating salaries); ③ Type Conversion (Forced conversion + automatic conversion, error-prone point: format issues when converting str to int); ④ Practical Exercise: Record personal information (name, age, salary) and print it
3. Operators and Expressions	① Arithmetic Operators (+/-/*//%** , cases: calculating shopping discounts, simplified version of salary tax calculation); ② Assignment Operators (=/:=, etc., pitfall to avoid: priority of chained assignment); ③ Comparison Operators (==/!=/>/<, case: judging whether a score is passing); ④ Logical Operators (and/or/not, case: judging whether the condition "adult and having income" is met); ⑤ Practical Exercise: Simple Calculator (implement addition, subtraction, multiplication and division)

Module Order	Core Content
4. Flow Control (Core Foundation)	① if-elif-else (cases: grade classification, leap year judgment); ② for loop (traversing strings/lists, case: batch printing names); ③ while loop (conditional loop, cases: countdown, number guessing game (1-100)); ④ Loop Control (break/continue, pitfall to avoid: indentation issues in nested loops); ⑤ Practical Exercise: Number Guessing Game (add fault tolerance mechanism to prevent crashes when non-numeric input is entered)
5. Function Basics	① Function Definition and Calling (def keyword, parameters, return values); ② Parameter Types (positional parameters, keyword parameters, default parameters, pitfall to avoid: parameter order); ③ Nested Functions (simple nesting, case: calculating complex formulas); ④ Anonymous Functions (lambda, case: simple calculations); ⑤ Practical Exercise: Encapsulate "score judgment function" and "area calculation function"
6. Data Structures (List/Tuple/Dictionary/Set)	① List (creation/addition/deletion/modification/query, case: shopping list management); ② Tuple (immutable feature, case: recording ID numbers/coordinates); ③ Dictionary (key-value pairs, case: student information management (name-score)); ④ Set (deduplication/intersection/union, case: filtering duplicate data); ⑤ Pitfalls to avoid: list index out of bounds, immutable dictionary keys; ⑥ Practical Exercise: Student Score Management System (addition/deletion/modification/query)
7. File Operations	① Opening/Closing Files (open function, with statement, pitfall to avoid: forgetting to close files); ② Text File Reading and Writing (read/readline/write, cases: reading score files, writing logs); ③ Basic CSV File Reading and Writing (case: batch importing student information); ④ Practical Exercise: Write student scores to a CSV file and read it

Module Order	Core Content
8. Exception Handling	① Concept of Exceptions (try-except-finally); ② Common Exceptions (ValueError/TypeError/FileNotFoundError, etc.); ③ Custom Exceptions (simple case); ④ Practical Exercise: Optimize the Number Guessing Game/Calculator (add exception capture to avoid crashes)
9. Basic Comprehensive Practical Project	① Project: Simple Address Book (implement addition/deletion/modification/query + file saving); ② Code Review: Sort out core knowledge points + common errors; ③ Assignment: Optimize the address book (add search function)

1.2 Code Files



https://github.com/XianZS/PythonLearning/tree/main/python_basic_learning

1.3 Learning Documents



https://github.com/XianZS/PythonLearning/tree/main/python_basic_learning/word

2.1 Python Advanced Learning Section

Module Order	Core Content
1. Advanced Functions	① Variable-Length Arguments (*args/**kwargs, case: batch processing an uncertain number of parameters); ② Decorators (basic principles + syntactic sugar, cases: function timing, log recording); ③ Generators (yield keyword, case: batch generating data to solve memory usage issues); ④ Iterators (iter/next, compare differences with generators); ⑤ Pitfalls to avoid: order of nested decorators, lazy evaluation of generators; ⑥ Practical Exercise: Optimize the previous address book project with decorators (add logs)
2. Object-Oriented Programming (OOP)	① Classes and Objects (definition/instantiation, cases: creating "Student class" and "Teacher class"); ② Encapsulation/Inheritance/Polymorphism (core features, case: Student class inherits from "Human class" and overrides methods); ③ Class Attributes and Instance Attributes (pitfall to avoid: attribute name conflicts); ④ Magic Methods (init/str/repr, etc., case: optimize the print output of classes); ⑤ Practical Exercise: Reconstruct the address book project with OOP (create Contact class to implement encapsulation)
3. Modularization and Package Management	① Module Import (import/from...import, pitfalls to avoid: circular import, module path issues); ② Custom Modules (split the address book project into multiple modules: business logic/file operations/exception handling); ③ Package Creation (role of init.py); ④ Virtual Environment (venv creation to solve dependency conflicts); ⑤ PyPI Release (simple demonstration, optional); ⑥ Practical Exercise: Split the address book into a modular project and create a virtual environment

Module Order	Core Content
4. Basic Data Processing (Connecting Standard Libraries/Third-Party Libraries)	① Advanced String Processing (basic regular expressions, re module, case: extracting phone numbers/emails from text); ② Date and Time Processing (datetime module, cases: log time formatting, calculating date differences); ③ Data Serialization (json module, case: convert address book data to JSON for saving); ④ Practical Exercise: Batch extract contact information from text and save as JSON
5. Basic Network Programming	① Basic HTTP Protocol (request/response, GET/POST methods); ② urllib Library Usage (case: simple web page text crawling); ③ requests Library (third-party, simplified requests, case: crawling weather data); ④ Pitfall to avoid: basic anti-crawling (User-Agent setting); ⑤ Practical Exercise: Crawl the weather of the local city and save it as text
6. Database Programming	① Basic SQLite (built-in database, no installation required, case: creating a student score database); ② Basic SQL Statements (addition/deletion/modification/query, CRUD); ③ Python Operation SQLite (sqlite3 module, case: storing address book data in the database); ④ MySQL Introduction (optional, pymysql library, case: batch inserting data); ⑤ Pitfalls to avoid: SQL injection, database connection closure; ⑥ Practical Exercise: Replace files with databases to optimize the address book project

Module Order	Core Content
7. Introduction to Concurrent Programming	① Threads and Processes (conceptual differences, GIL Global Interpreter Lock); ② threading Module (thread creation, case: multi-threaded image downloading); ③ multiprocessing Module (process creation, case: multi-process data processing); ④ Pitfalls to avoid: thread safety issues, impact of GIL on CPU-intensive tasks; ⑤ Practical Exercise: Multi-threaded crawling of weather data from multiple web pages
8. Advanced Comprehensive Practical Projects (2 Projects, Covering Multiple Modules)	① Project 1: Simple Web Crawler + Data Visualization (crawl commodity prices from a platform, plot with matplotlib, covering network requests, data processing, visualization); ② Project 2: Student Score Management System (upgraded version, covering OOP, database, modularization, exception handling); ③ Each project is divided into 3 videos: "Requirements Analysis → Code Implementation → Optimization Review"
9. Advanced Stage Review and Direction Guidance	① Core Knowledge Point Sorting (mind map format); ② Summary of Common Errors and Solutions; ③ Follow-up Direction Guidance (data analysis/web crawling/automation/backend development)

2.2 Code Files



https://github.com/XianZS/PythonLearning/tree/main/python_advance_learning

2.3 Learning Documents



https://github.com/XianZS/PythonLearning/tree/main/python_advance_learning/word

3.1 Python Standard Library Learning

Section

Usage Scenario	Standard Library Modules	Main Usage Description
File and Directory Operations	os, os.path, shutil, pathlib, tempfile	Provide operating system interfaces for file/directory creation, deletion, copying, moving, path processing, and temporary file management.
Data Processing and Formats	json, xml, csv, configparser, pickle, sqlite3, collections, itertools	Process JSON/XML/CSV data, parse configuration files, serialize objects, operate SQLite databases, and provide efficient container/iterator tools.
System Interaction and Processes	sys, subprocess, argparse, logging, platform	Access command-line arguments, start subprocesses, record logs, obtain system information, and interact with the Python interpreter.
Text Processing	re, string, difflib, textwrap	Regular expression matching, string operations, text difference comparison, automatic line wrapping/filling and other text processing functions.
Date and Time	datetime, time, calendar	Process dates, times, and calendars, supporting time calculation, formatted output, and time zone conversion.

Usage Scenario	Standard Library Modules	Main Usage Description
Mathematics and Scientific Computing	<code>math</code> , <code>cmath</code> , <code>random</code> , <code>statistics</code> , <code>fractions</code> , <code>decimal</code>	Provide mathematical functions, complex number operations, random number generation, statistical calculations, and high-precision fraction/decimal operations.
Concurrency and Asynchrony	<code>threading</code> , <code>multiprocessing</code> , <code>concurrent.futures</code> , <code>asyncio</code> , <code>queue</code>	Implement multi-threading, multi-processing, thread pools/process pools, and asynchronous IO programming, supporting task queues and concurrency control.
Debugging and Testing	<code>pdb</code> , <code>unittest</code> , <code>doctest</code> , <code>traceback</code>	Code debugging, unit testing, document testing, and exception stack information acquisition, facilitating development and testing processes.
Network Communication	<code>socket</code> , <code>urllib</code> , <code>http</code> , <code>ftplib</code> , <code>smtp</code>	Implement low-level network communication, HTTP requests, FTP file transfer, email sending and other network protocol functions.
Encryption and Security	<code>hashlib</code> , <code>hmac</code> , <code>secrets</code>	Provide hash algorithms, message authentication codes, secure random number generation, and SSL/TLS encrypted communication support.

3.2 Code Files



https://github.com/XianZS/PythonLearning/tree/main/python_stdlib_learning

3.3 Learning Documents



https://github.com/XianZS/PythonLearning/tree/main/python_stdlib_learning/word



Project Star Icon (项目星标)



Star History

